

This code also does not need any explanation, as it is similar to the other undirected simple graphs generated earlier in this series. Upon execution of the program `'disgraph9.sagews'`, we get the graph called *Disgraph_9* shown in Figure 2. This graph shows the rail connectivity between Delhi, Mumbai, Kolkata, Chennai, Kavaratti (the capital of the union territory of Lakshadweep), and Port Blair (the capital of the union territory of the Andaman and Nicobar Islands). Keep in mind that *Disgraph_9* only represents connectivity between places and not the actual geographical location of these places on the map of India. We all know that a person can travel by train between Delhi, Mumbai, Kolkata, and Chennai. However, since Kavaratti and Port Blair are located on islands, there is no rail connectivity to those cities from other cities in India. In situations like this, disconnected graphs are very useful for capturing essential data from real-world scenarios.

Let us also familiarise ourselves with some connectivity-related methods available in SageMath. To do that, add the following lines of code at the end of the program called `'disgraph9.sagews'`.

```
G.is_connected( )
G.connected_components_number( )
G.connected_components(sort=True)
G.connected_components_sizes( )
G.all_paths('Delhi', 'Mumbai')
G.all_paths('Delhi', 'Port Blair')
```

Upon execution of the modified program, the following additional output will be displayed on the screen.

```
False
3
[['Chennai', 'Delhi', 'Kolkata', 'Mumbai'], ['Port Blair'],
 ['Kavaratti']]
[4, 1, 1]
[['Delhi', 'Mumbai'], ['Delhi', 'Kolkata', 'Chennai',
 'Mumbai']]
[]
```

Now, let us compare the code and the corresponding output for better understanding. The method `G.is_connected()` checks if the graph *G* is connected. Since the graph *Disgraph_9* is not connected, the answer is given as *False*. The method `G.connected_components_number()` returns the number of connected components in the graph *G*. Since Delhi, Mumbai, Kolkata, and Chennai are connected with each other, they form one connected component. Kavaratti and Port Blair are located on islands, forming two more components. Hence, the answer is 3. The method `G.connected_components(sort=True)` returns the connected

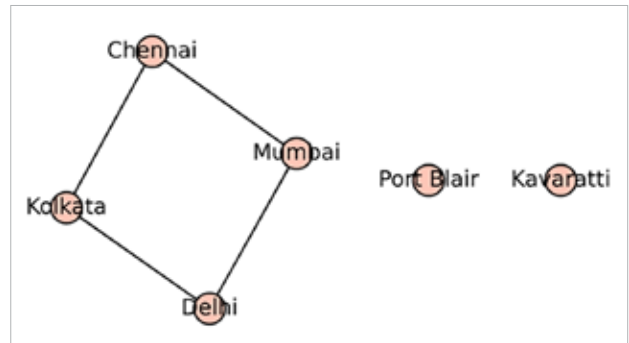


Figure 2: The disconnected graph *Disgraph_9*

components of the graph *G*. The parameter `sort=True` ensures that the vertices inside each component are sorted according to the default ordering. From the output, we can see that the list has three elements, each containing the names of the connected components.

The method `G.connected_components_sizes()` returns the sizes of the connected components in the graph *G*. The size indicates the number of vertices in a particular connected component. Comparing with the previous output, we can see that the output `[4, 1, 1]` is correct. The method call `G.all_paths('Delhi', 'Mumbai')` returns all possible paths between the vertices `'Delhi'` and `'Mumbai'` in the graph *G*. The list shows the direct path from `'Delhi'` to `'Mumbai'` as well as the indirect path through vertices `'Kolkata'` and `'Chennai'`. Similarly, the method call `G.all_paths('Delhi', 'Port Blair')` returns all paths between vertices `'Delhi'` and `'Port Blair'` if they exist. Since there is no path between `'Delhi'` and `'Port Blair'`, this method call returns an empty list, as you can observe in the output above.

Now, let us move on to the next type of graph, the directed graph (also known as a digraph). But why do we need directed graphs? We already have undirected simple graphs and multigraphs, along with a number of powerful methods to elicit information from them. Well, the reason is that undirected graphs (simple and multigraphs) can only represent symmetric relationships between two entities or objects. Let us consider the example we discussed in the second article in this series, available on the OSFY portal at <https://www.opensourceforu.com/2024/03/sagemath-demystifying-graphs/>. In the example, we used an undirected graph to capture the friendship between six people. An edge between a pair of vertices indicates that they are friends with each other. However, human relationships do not always work this way. What if I am in love with somebody and she doesn't care about me at all? Well, an undirected graph cannot represent such a relationship effectively.

Let us use three characters from the world-famous play *'No Exit'*, written by Jean-Paul Sartre, a renowned philosopher and writer (a Nobel Laureate in literature and the